PART A — QUESTIONS

AF5-001
Question:
Solve: $7x + 6 = 48$
A) 5
B) 6
C) 7
D) 8

AF5-002
Question:
If $5n - 12 = 28$, what is n?
A) 6
B) 7
C) 8
D) 9

AF5-003
Question:
Simplify: $4(3x + 2) - 5x$
A) $7x + 8$
B) $12x + 8$
C) $7x + 2$
D) $17x + 8$

AF5-004
Question:
A company charges $16 per hour plus a $25 setup fee. Which expression represents the total charge for h hours?
A) $16h + 25$
B) $25h + 16$
C) $41h$
D) $16 + 25h$

AF5-005
Question:
If $f(x) = 3x + 8$, what is $f(9)$?
A) 27
B) 30
C) 35
D) 37

AF5-006
Question:
Which ordered pair satisfies the equation $y = 3x + 4$?
A) (1, 6)
B) (2, 10)
C) (3, 12)
D) (4, 18)

AF5-007
Question:
Solve: 6(x + 2) = 54
A) 7
B) 8
C) 9
D) 10

AF5-008
Question:
A pattern begins: 5, 12, 19, 26, ... What is the next number?
A) 31
B) 32
C) 33
D) 34

AF5-009
Question:
Solve: x/9 = 5
A) 14
B) 36
C) 45
D) 54

AF5-010
Question:
A line has a slope of 4 and crosses the y-axis at -7. Which equation represents the line?
A) y = -7x + 4
B) y = 4x - 7
C) y = 7x - 4
D) y = -4x + 7

AF5-011
Question:
If 6 cases contain 90 units, how many units are in 10 cases at the same rate?
A) 120
B) 135
C) 150
D) 180

AF5-012
Question:
Solve: 30 - 5x = 10
A) 2
B) 3
C) 4
D) 5

AF5-013
Question:
Which expression is equivalent to 16a - 9a + 11?

A) 7a + 11
B) 25a + 11
C) 7a - 11
D) 16a + 2

AF5-014
Question:
A tank starts with 60 gallons and is filled at a rate of 8 gallons per minute. Which equation gives the total amount A after m minutes?
A) A = 60m + 8
B) A = 8m + 60
C) A = 68m
D) A = 60 - 8m

AF5-015
Question:
If y = -4x + 15, what is y when x = 2?
A) 7
B) 8
C) 17
D) 23

AF5-016
Question:
Solve: 10x - 14 = 4x + 22
A) 4
B) 5
C) 6
D) 8

AF5-017
Question:
Which value of x makes 2x + 9 less than 23?
A) 5
B) 6
C) 7
D) 8

AF5-018
Question:
A table shows a linear relationship.

x: 0, 1, 2, 3
y: 6, 11, 16, 21

What is the rule?
A) y = 6x + 5
B) y = 5x + 6
C) y = 11x - 5
D) y = x + 6

AF5-019
Question:
Solve: 7(x - 1) = 3x + 17
A) 4
B) 5
C) 6
D) 7

AF5-020
Question:
If c = -6 and d = 4, what is 3d - c?
A) 6
B) 12
C) 18
D) -18

AF5-021
Question:
Which point is located in Quadrant III?
A) (4, -5)
B) (-4, 5)
C) (-4, -5)
D) (4, 5)

AF5-022
Question:
A worker packs 35 items in 5 minutes. At the same rate, how many items can the worker pack in 8 minutes?
A) 48
B) 52
C) 56
D) 63

AF5-023
Question:
Solve for b: M = 4b + 12
A) b = M - 12
B) b = (M - 12)/4
C) b = 4(M - 12)
D) b = M/4 + 12

AF5-024
Question:
What is the slope of a line that goes through (2, 8) and (6, 20)?
A) 2
B) 3
C) 4
D) 12

AF5-025
Question:

Simplify: $11x - 6 - 4x + 9$
A) $7x + 3$
B) $15x + 3$
C) $7x - 15$
D) $15x - 15$

AF5-026
Question:
If $g(x) = x^2 - 2x$, what is $g(7)$?
A) 35
B) 42
C) 49
D) 63

AF5-027
Question:
A sequence follows the rule “multiply by 2, then add 5.” If the first term is 6, what is the second term?
A) 12
B) 15
C) 17
D) 22

AF5-028
Question:
Solve: $5(x + 1) - 4 = 31$
A) 5
B) 6
C) 7
D) 8

AF5-029
Question:
A line crosses the y-axis at 8 and falls 3 units for every 1 unit to the right. Which equation represents the line?
A) $y = 3x + 8$
B) $y = -3x + 8$
C) $y = 8x - 3$
D) $y = -8x + 3$

AF5-030
Question:
If $7/10 = x/40$, what is x?
A) 21
B) 24
C) 28
D) 35

AF5-031
Question:
Solve: $-7x + 14 = -35$

A) -7
B) -3
C) 3
D) 7

AF5-032
Question:
A storage shelf starts with 180 parts. Each order removes 15 parts. Which expression gives the number of parts left after o orders?
A) 180 + 15o
B) 180 - 15o
C) 15o - 180
D) 180 ÷ 15o

AF5-033
Question:
Which equation has x = 10 as its solution?
A) 3x + 4 = 32
B) 5x - 8 = 42
C) x/2 + 3 = 9
D) 4x - 5 = 30

PART B — ANSWER KEY + EXPLANATIONS

AF5-001 — Correct: B — Explanation:
Solve 7x + 6 = 48. Subtract 6 from both sides: 7x = 42. Divide by 7: x = 6. The most common mistake is subtracting 6 correctly but forgetting to divide by 7.

AF5-002 — Correct: C — Explanation:
Start with 5n - 12 = 28. Add 12 to both sides: 5n = 40. Divide by 5: n = 8. A common mistake is subtracting 12 instead of adding it.

AF5-003 — Correct: A — Explanation:
Distribute first: 4(3x + 2) = 12x + 8. Then subtract 5x: 12x - 5x + 8 = 7x + 8. A common mistake is forgetting to multiply 2 by 4.

AF5-004 — Correct: A — Explanation:
The company charges $16 for each hour, so that part is 16h. The $25 setup fee is added one time. The total charge is 16h + 25. A common mistake is treating the setup fee as the hourly rate.

AF5-005 — Correct: C — Explanation:
Substitute 9 for x: f(9) = 3(9) + 8 = 27 + 8 = 35. A common mistake is multiplying correctly but forgetting to add 8.

AF5-006 — Correct: B — Explanation:
Test the choices in y = 3x + 4. For (2, 10), 3(2) + 4 = 6 + 4 = 10, which matches the y-value. A common mistake is choosing a point that is close but does not make both sides equal.

AF5-007 — Correct: A — Explanation:
Solve 6(x + 2) = 54. Divide both sides by 6: x + 2 = 9. Subtract 2: x = 7. A common mistake is subtracting 2 before undoing the multiplication by 6.

AF5-008 — Correct: C — Explanation:
The pattern increases by 7 each time: 5, 12, 19, 26. Add 7 to 26: 33. A common mistake is assuming the pattern increases by 6 because 26 is close to 32.

AF5-009 — Correct: C — Explanation:
$x/9 = 5$ means x divided by 9 equals 5. Multiply both sides by 9: $x = 45$. A common mistake is adding 9 and 5 instead of multiplying.

AF5-010 — Correct: B — Explanation:
Slope-intercept form is $y = mx + b$. The slope is 4 and the y-intercept is -7, so the equation is $y = 4x - 7$. A common mistake is switching the slope and y-intercept.

AF5-011 — Correct: C — Explanation:
Find the unit rate: 90 units ÷ 6 cases = 15 units per case. For 10 cases: $15 \times 10 = 150$. A common mistake is adding 4 cases and 4 units instead of using the rate.

AF5-012 — Correct: C — Explanation:
Solve $30 - 5x = 10$. Subtract 30 from both sides: $-5x = -20$. Divide by -5: $x = 4$. A common mistake is losing the negative sign after subtracting 30.

AF5-013 — Correct: A — Explanation:
Combine like terms: $16a - 9a = 7a$. The constant 11 stays the same. The expression becomes $7a + 11$. A common mistake is combining the constant with the variable terms.

AF5-014 — Correct: B — Explanation:
The tank starts with 60 gallons. It gains 8 gallons each minute, so after m minutes the amount is $A = 8m + 60$. A common mistake is making 60 the rate instead of the starting amount.

AF5-015 — Correct: A — Explanation:
Substitute $x = 2$ into $y = -4x + 15$: $y = -4(2) + 15 = -8 + 15 = 7$. A common mistake is treating $-4(2)$ as positive 8.

AF5-016 — Correct: C — Explanation:
Solve $10x - 14 = 4x + 22$. Subtract 4x: $6x - 14 = 22$. Add 14: $6x = 36$. Divide by 6: $x = 6$. A common mistake is adding 14 to only one side.

AF5-017 — Correct: A — Explanation:
Solve $2x + 9 < 23$. Subtract 9: $2x < 14$. Divide by 2: $x < 7$. The only listed value less than 7 is 5. A common mistake is choosing 7, but $2(7) + 9 = 23$, not less than 23.

AF5-018 — Correct: B — Explanation:
The y-values increase by 5 each time x increases by 1, so the slope is 5. When $x = 0$, $y = 6$, so the y-intercept is 6. The rule is $y = 5x + 6$. A common mistake is using 6 as the slope because it is the first y-value.

AF5-019 — Correct: C — Explanation:
Distribute: $7x - 7 = 3x + 17$. Subtract 3x: $4x - 7 = 17$. Add 7: $4x = 24$. Divide by 4: $x = 6$. A common mistake is forgetting to multiply -1 by 7.

AF5-020 — Correct: C — Explanation:
Substitute $c = -6$ and $d = 4$: $3d - c = 3(4) - (-6) = 12 + 6 = 18$. A common mistake is treating subtracting -6 as subtracting 6.

AF5-021 — Correct: C — Explanation:
Quadrant III has a negative x-value and a negative y-value. The point (-4, -5) fits that pattern. A common mistake is choosing (-4, 5), which is Quadrant II.

AF5-022 — Correct: C — Explanation:
35 items in 5 minutes means $35 \div 5 = 7$ items per minute. In 8 minutes: $7 \times 8 = 56$. A common mistake is multiplying 35 by 8 without finding the rate first.

AF5-023 — Correct: B — Explanation:
Start with $M = 4b + 12$. Subtract 12 from both sides: $M - 12 = 4b$. Divide by 4: $b = (M - 12)/4$. A common mistake is dividing M by 4 first and then subtracting 12.

AF5-024 — Correct: B — Explanation:
Slope equals change in y divided by change in x. From (2, 8) to (6, 20), y increases by 12 and x increases by 4. The slope is $12 \div 4 = 3$. A common mistake is using 12 as the slope without dividing by the change in x.

AF5-025 — Correct: A — Explanation:
Combine like terms: $11x - 4x = 7x$, and $-6 + 9 = 3$. The simplified expression is $7x + 3$. A common mistake is writing $7x - 3$ because the positive 9 is missed.

AF5-026 — Correct: A — Explanation:
Substitute 7 for x: $g(7) = 7^2 - 2(7) = 49 - 14 = 35$. A common mistake is calculating 7^2 as 14 instead of 49.

AF5-027 — Correct: C — Explanation:
Start with 6. Multiply by 2: 12. Then add 5: 17. A common mistake is adding 5 first and then multiplying.

AF5-028 — Correct: C — Explanation:
Solve $5(x + 1) - 4 = 31$. Distribute: $5x + 5 - 4 = 31$. Combine: $5x + 1 = 31$. Subtract 1: $5x = 30$. Divide by 5: $x = 6$. Wait, this gives $x = 6$, so the correct answer should be B, not C. This item must be corrected before use.

AF5-029 — Correct: B — Explanation:
“Falls 3 units for every 1 unit to the right” means the slope is -3. The line crosses the y-axis at 8, so the equation is $y = -3x + 8$. A common mistake is using a positive slope because the number 3 is given.

AF5-030 — Correct: C — Explanation:
Use the proportion $7/10 = x/40$. Since $10 \times 4 = 40$, multiply 7 by 4: $x = 28$. A common mistake is adding 30 to the numerator because 10 becomes 40.

AF5-031 — Correct: D — Explanation:
Solve $-7x + 14 = -35$. Subtract 14 from both sides: $-7x = -49$. Divide by -7: $x = 7$. A common mistake is giving -7 because both sides contain negative numbers.

AF5-032 — Correct: B — Explanation:
The shelf starts with 180 parts. Each order removes 15 parts, so o orders remove $15o$ parts. The number left is $180 - 15o$. A common mistake is adding $15o$ even though parts are being removed.

AF5-033 — Correct: B — Explanation:
Test $x = 10$. In choice B, $5(10) - 8 = 50 - 8 = 42$, so the equation is true. A common mistake is choosing a nearby equation without checking both sides exactly.